

When a "Happy" Worker Is Really a "Productive" Worker

A Review and Further Refinement of the Happy-Productive Worker Thesis

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For decades, organizational scientists and practitioners alike have been fascinated by the happy-productive worker thesis. According to this hypothesis, happy employees exhibit higher levels of job-related performance behaviors than do unhappy employees. However, despite years of research, support for the happy-productive worker thesis remains equivocal. These ambiguous findings result from the variety of ways in which happiness has been operationalized. Researchers have operationalized happiness as job satisfaction, as the presence of positive affect, as the absence of negative affect, as the lack of emotional exhaustion, and as psychological well-being. Some of these measures exhibit appreciable associations with job performance, others do not. The circumplex framework is offered as a potentially useful taxonomy for researchers interested in better understanding and promoting a happy and productive workforce.

Every one of these hundreds of millions of human beings is in some form seeking happiness.

—Hubert George Wells, *The Outline of History*, 1920

Similar to Diogenes the Cynic and his quest to find an honest man, many researchers and practitioners alike have to come to consider the happy-productive worker thesis as a holy grail of the organizational sciences (Landy, 1985). Yet decades of dialogue have not resolved the question, and many organizational scientists remain skeptical about the relation between happiness and job performance (Staw, 1986; Wright & Staw, 1999). Issues of worker health and happiness have arguably never been more important than they are as

we enter the new millennium. For instance, former American Psychological Association President Martin Seligman (as cited in Mattox, 1999) noted that an "epidemic" of clinical depression currently exists in the United States. To be more specific, research related to the happy-productive worker thesis assumes even greater relevance when we consider Myers and Diener's (1997) claim that only a minority of Americans can be considered happy. Whether one agrees with Myers and Diener's assessment, the fact is that roughly 17 million Americans are currently taking Prozac, with millions more taking other forms of antidepressant medications (Wright, 1999). Although there are

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many causes of unhappiness, some are undoubtedly the result of work-related issues.

In this article, we reexamine the happy-productive worker thesis. We propose that the equivocal findings to date are largely due to the inconsistent manner in which happiness has been understood. To be more specific, although there is reasonable consensus in regard to the conceptual definition of happiness (Diener, 1984), there is less agreement as to the operational definition. To illustrate this fact, in the pages that follow, we identify a variety of ways in which happiness has been operationalized in organizational research. As we shall demonstrate, some of these approaches are more robust than others in predicting performance. Despite this caveat, our general conclusion is an optimistic one for management scholars and practitioners alike: Happiness promotes higher job performance.

Why Happiness Should Be Related to Job Performance

Diener (1984) noted that virtually all scientific approaches to happiness converge around three defining phenomenon. First, happiness is a subjective experience (Diener, 1994; Diener, Sandvik, Seidlitz, & Diener, 1993; Parducci, 1995). People are happy to the extent that they believe themselves to be happy. For this reason, happiness involves some type of judgment as to the pluses and minuses of one's life (Parducci, 1995). Second, happiness includes both the relative presence of positive emotions and the relative absence of negative emotions (Argyle, 1987; Diener & Larsen, 1993; Michalos, 1985; Warr, 1987, 1990). Third, happiness is a global judgment. It refers to one's life as a whole. As the term is used here, happiness is an overall evaluation that exhibits some measure of stability over time (Diener, 1994; Diener, Suh, Lucas, & Smith, 1999; Myers, 1993; Veenhoven, 1988), though it is strongly influenced by environmental events (Pavot, Diener, Colvin, & Sandvik,

1991; Sandvik, Diener, Seidlitz, 1993) and responsive to therapeutic interventions (Seligman, 1994, 1995). Taken together, it is generally accepted that happiness refers to a subjective and global judgment that one is experiencing a good deal of positive emotion and relatively little negative emotion.

All things being equal, few people would prefer to be unhappy when they could be happy. In fact, Sullivan (1989) argued that seeking pleasure and avoiding pain is fundamental to human motivation. Given the worth that most people place on happiness and the fact that only some people have enough of it, we can begin to understand the relation between happiness and job performance. When one looks across a group of people, happiness can be seen as a valuable, though sometimes scarce, resource (Hobfoll, 1998; Myers & Diener, 1997). This inherent value placed on happiness, coupled with its occasional scarcity, underscores the importance of conserving or maintaining happiness whenever possible (Hobfoll, 1988, 1989, 1998).

The necessity for resource maintenance places happy and unhappy people in different psychological situations. When an unhappy person goes to work, he or she needs to protect and defend his or her relatively limited reserve of happiness. One is less likely to take risks when something precious is in short supply. For this reason, unhappy people (a) are sensitive to threats in their work environment, (b) are defensive and cautious around their coworkers, and (c) are less optimistic and confident. Happy people, on the other hand, have a more plentiful stock of a valuable resource. When at work, they can more afford to take risks that could potentially increase their supply of happiness. For this reason, happy people (a) are more sensitive to opportunities in their work environments, (b) are more outgoing and helpful to their coworkers, and (c) are more

optimistic and confident. According to this resource maintenance model, the need to conserve their limited resource supply can lead unhappy people to be poorer performers, whereas the flexibility afforded by their more bountiful resource reserve can improve the performance of happy persons. These possibilities are discussed more thoroughly in the paragraphs that follow.

When something precious is also rare, individuals are more sensitive to cues that signal a threat (Hobfoll, 1989; Lee & Ashforth, 1996). On the other hand, individuals can afford less worry when a resource is more abundant. As a result, happy people are more sensitive to positive events, whereas unhappy people are more sensitive to negative events. For example, experimental research by Seidlitz and Diener (1993) and Seidlitz, Wyer, and Diener (1997) found that, when compared with their happier counterparts, those who were low on well-being were more likely to encode an ambiguous event as threatening. Likewise, another experiment by Larsen and Ketelaar (1989) found that unfavorable feedback was more hurtful to those who were prone to negative emotions and less hurtful to those who were prone to positive emotions. On the other hand, favorable feedback yielded larger benefits for those who were predisposed to positive emotions while yielding smaller benefits to those predisposed to negative emotions (for more evidence, see Derryberry & Read, 1994; Larsen & Ketelaar, 1991; Rusting & Larsen, 1997). These effects persist over time because of the manner in which happy and unhappy people recall events. Happy people tend to remember favorable events, whereas unhappy people tend to remember unfavorable ones (Seidlitz & Diener, 1993; Seidlitz et al., 1997). This tendency to emphasize the negative aspects of work life is likely to have deleterious consequences for job performance.

Interacting with other people is simultaneously fraught with emotional peril but filled with opportunities. Unhappy people,

who are more cautious and protective in social situations, sometimes incline toward introversion and shyness (Argyle, 1987). Happy people tend to be more outgoing and extroverted (Diener, Sandvik, Pavot, & Fujita, 1992; Headey & Wearing, 1988; Myers & Diener, 1995). At times, unhappy people can even become acrimonious. In one longitudinal study, Bolger and Schilling (1991) found that people who were prone to negative emotion were apt to use contentious interpersonal tactics, thereby provoking the ire of coworkers. It is no surprise that unhappy people report feeling less coworker and supervisory support than do their more happy counterparts (Staw, Sutton, & Pelled, 1994). All of these findings suggest that, generally speaking, happy people would perform better on the wide range of jobs that require social interaction.

Accentuating negative events and minimizing social contact eventually takes its toll. Relative to their happier coworkers, unhappy people are likely to have lower self-esteem (Diener et al., 1999; Myers, 1993). Likewise, unhappy people see themselves as having less control over events in their lives and are less optimistic about the future (Dember & Brooks, 1989; Seligman, 1991). This sense of demoralization may make unhappy people less proactive (Argyle, 1987) and more prone to stress symptoms (Myers & Diener, 1995).

If happiness should conceptually be related to job performance, then why are the connections viewed by some scholars as tenuous? In our view, the problem has to do with the diverse manner in which happiness has been operationalized. When testing the happy-productive worker thesis, both scholars and practitioners have primarily treated happiness in four ways: as job satisfaction, as the profile of positive and negative affectivity, as the lack of emotional exhaustion, and as psychological well-being. The results of these four operationalizations are not fully con-

sistent. In the remainder of this article, we examine tests of the happy-productive worker thesis to workplace happiness using each of these approaches.

Happiness as Job Satisfaction: More Than Meets the Eye?

Historically, a common means of testing the happy-productive worker thesis was by operationalizing happiness as job satisfaction. This approach was favored by early researchers, such as Herzberg, Mausner, and Snyderman (1959); Houser (1927); and Roethlisberger and Dickson (1939). Although contemporary scholars distinguish between satisfaction with a particular job and happiness with life as a whole (e.g., Diener et al., 1999; Judge, 1992; Judge & Locke, 1993), historical convention has sometimes led the happy-productive worker thesis to be operationalized as the relation between job satisfaction and job performance. For example, Judge, Thoresen, Bono, and Patton (1998, p. 3) opened their recent meta-analysis of the job satisfaction-performance relation with the question: "Is a happy worker a productive one?"

With this in mind, several reviews have provided excellent summaries of the satisfaction-performance relation. For example, reviewing the literature to date, Brayfield and Crockett (1955, p. 421) determined that "satisfaction ... need not imply strong motivation to outstanding performance." Similar conclusions were reached in a later review by Vroom (1964), who found that the median association between these two variables was a rather modest $+0.14$. Finally, a meta-analytic review by Iaffaldano and Muchinsky (1985) obtained a mean corrected correlation of $+0.17$. Together, these reviews generated a good deal of skepticism regarding the relation between job satisfaction and performance (e.g., Cherrington, Reitz, & Scott, 1971; Fisher, 1980; Locke, 1976; Podsakoff & Williams, 1986; Staw, 1986).

However, other evidence has called these modest conclusions into question.

In a recent review, Judge et al. (1998) argued that the earlier meta-analytic review by Iaffaldano and Muchinsky (1985) may have produced overly conservative results. In fact, a meta-analysis by Petty, McGee, and Cavender (1984) placed the relation at $+0.31$. In their meta-analysis, Judge and his colleagues estimated the association to be roughly $+0.30$, though they observed some important moderators. For instance, they found some job satisfaction measures were more closely related to performance than others, ranging from an aggregated correlation of $+0.06$ for research measuring job satisfaction with the GM Faces Scale to an aggregated correlation of $+0.51$ for research using the Hoppock Job Satisfaction Blank to measure job satisfaction. Judge et al. also reported considerable differences across occupations, with the job satisfaction-performance correlation being more robust in high-complexity jobs. Finally, their meta-analytic results demonstrated that measures of overall job satisfaction display higher correlations than facet measures of job satisfaction. Despite the variability in the job satisfaction-performance relationship, many of the estimates are certainly large enough to have sizable financial consequences for work organizations (Cascio, 1991). As a result, some organizational scholars have become more sanguine about the prospect of finding a useful relation between job satisfaction and performance (e.g., Brief, 1998; Judge, Hanisch, & Drankoski, 1995; Podsakoff & Williams, 1986).

Of course, the positive associations obtained by Judge et al. (1998) and Petty et al. (1984) do not address the issue of causality. There are at least three causal relations between job satisfaction and performance: (a) Job satisfaction may cause job performance, (b) job performance may cause job satisfaction, and (c) both perfor-

mance and satisfaction may be caused by a variety of third variables. During the 1960s and 70s, several studies investigated these sundry causal paths, and at least nominal evidence was obtained for each (for reviews, see Lawler & Porter, 1967; Nord, 1976; Schwab & Cummings, 1970). Unfortunately, as Judge et al. (1998) observed, the general skepticism regarding the job satisfaction–performance relation led to a decline in research. As a consequence, the causal linkage between these two variables, if any, remains a worthwhile topic for future research.

Affective Dispositional Approaches to [Un]Happiness: [Un]Happiness as Positive Affectivity, Negative Affectivity, Emotional Exhaustion, and Well-Being

An Overview and Organizing Taxonomy

Much of the confusion underlying the term *happiness* stems from the overall structure of dispositional affect. Because happiness refers to emotions, a clear “map” of emotion terms would enhance conceptual precision. This need to dimensionalize emotional well-being has long been recognized (Larsen & Diener, 1992). In the second century, Galen devised a four-fold typology (e.g., melancholic, choleric, sanguine, and phlegmatic) to describe various differences in emotions. Much later, Kant expanded on Galen’s work in his treatise *Anthropologie*. Still later, Wundt transformed Galen and Kant’s work into a pleasantness–unpleasantness interpretation of emotion. More recently, Russell and his colleagues (Russell, 1978, 1979, 1980; Russell, Weiss, & Mendelsohn, 1989), using multidimensional scaling techniques, found that emotion terms can be organized into a roughly circular structure called a circumplex. An idealized circumplex structure is displayed in Figure 1. This circumplex structure has been found to be quite robust, as it general-

izes to both adults and children (Russell & Ridgeway, 1983) as well as across a variety of cultures (Russell, Lewicka, & Nitt, 1989) and different analytic methods (Meyer & Shack, 1989). As a result, the circumplex structure is widely accepted (e.g., Larsen & Diener, 1992; Mayer & Gaschke, 1988), though the circumplex is not without problems of its own.

One concern with the circumplex is that it is often impractical to apply. Because the model represents emotions in two-dimensional space, it is not clear how one would use the entire model as a predictor of organizational behaviors. For this reason, researchers simplify the circumplex model by factor analyzing the emotion terms. Given the circular structure, different factor solutions could potentially provide an equally good fit to the model. For example, consider the solid vertical and horizontal axes in Figure 1. These two factors result from an unrotated factor model. The vertical factor designates “hedonic tone” or “pleasantness–unpleasantness.” It is anchored by such adjectives as “joyous” on the high end and “sad” on the low. Thus, this dimension is often interpreted as psychological well-being (e.g., Myers, 1993). On the other hand, the horizontal axis refers to the level of activation, or “affect intensity,” of emotional experience (Larsen & Diener, 1992). In the organizational sciences, this arousal dimension has sometimes been understood as emotional exhaustion. For example, Wright and Bonett (1997a, 1997b) maintained that high emotional exhaustion can be treated as low emotional activation–intensity. This makes sense insofar as exhaustion is defined as a numb state of emotional detachment (Maslach, 1982; Maslach & Leiter, 1997), though we shall qualify this observation later.

When the primary dimensional factors are rotated 45 degrees, two new dimensions emerge: positive and negative affectivity. These are designated in Figure 1 by the two dashed lines. The positive affectivity (PA)

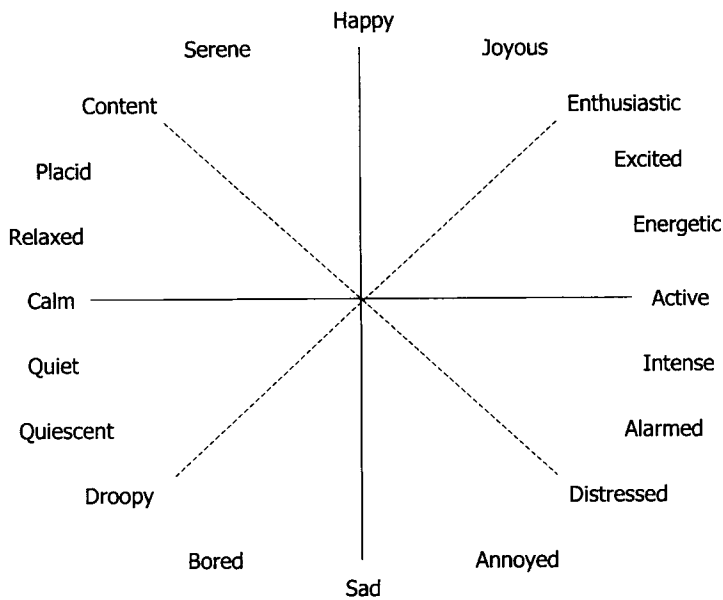


Figure 1. An idealized representation of the circumplex model of dispositional happiness. The solid lines refer to the unrotated factor solution, and the dashed lines refer to the rotated solution.

dimension begins in the upper right quadrant of Figure 1 and runs diagonally to the lower left quadrant. High PAs tend to report experiencing positive emotions, such as “enthusiastic” and “excited.” At the low end, PA is best characterized by the absence of positive emotions, such as “bored.” In the lower right quadrant of Figure 1, and running to the upper left quadrant, is the negative affectivity (NA) dimension. High NAs report emotions that are negative, such as “alarmed” and “distressed.” Low NAs tend to endorse the absence of negative emotions. For example, a low NA might indicate a tendency to be “placid” or “relaxed.” There is considerable evidence that the NA–PA solution is a useful one (e.g., Brief, Butcher, & Roberson, 1995; Brief & Roberson, 1989; Burke, Brief, George, Roberson, & Webster, 1989; George, 1996; Watson, 1988; Watson & Clark, 1984; Watson, Clark, & Tellegen, 1988; Watson & Tellegen, 1985). Moreover, NA and PA show up across cultures (Watson, Clark, & Tellegen, 1984) and when both self- and peer ratings are examined.

These are the most widely accepted organizational perspectives on dispositional affect. More important, each of these approaches shares a common origin in the circumplex model. In particular, they result from two different rotational solutions and, as such, are different representations of the same underlying reality. We will now consider how each of these perspectives relates to job performance.

The Rotated Solution: Positive and Negative Affectivity

Happiness as positive and negative affectivity. According to Diener (1994; Diener et al., 1999), one way to conceptualize happiness is as the presence of dispositional positive affect (PA) and the absence of dispositional negative affect (NA). This approach is also evident in the work of Watson and Clark (1984) and Costa and McCrae (1980). As discussed by George (1992, 1996) and Judge (1992), there is now compelling evidence that PA and NA predict such work attitudes as job satisfaction

and organizational commitment (e.g., Cropanzano, James, & Konovsky, 1993; Levin & Stokes, 1989; Watson & Slack, 1993). NA also seems to be related to reports of job stress and strain (Brief, Burke, George, Robinson, & Webster, 1988; Burke, Brief, & George, 1993; Watson & Pennebaker, 1989). In addition, NA has been positively related to voluntary employee turnover (Wright & Cropanzano, 1998).

Given the resource maintenance model we presented earlier, one would expect PA to exhibit a positive association to job performance, whereas NA should exhibit a negative association. However, the evidence is equivocal. Wright and Cropanzano (1998) were unable to find a relation between either NA or PA and performance. Null relations were also reported by Wright and Staw (1999). However, Cropanzano et al. (1993) did obtain some significant interactions. In their Study 1, they found a PA by tenure interaction, such that PA was related to performance for employees who had high tenure but not for those with low tenure. Furthermore, they obtained an NA by tenure interaction in their Study 2. To be more specific, there was a relation between NA and performance for those who were low in tenure but not for those who were high. Unfortunately, these interactions were obtained post hoc and have yet to be replicated.

Part of the reason for these disappointing results may have less to do with dispositional affectivity and more to do with measurement. For example, Cropanzano and his colleagues used the respected Positive and Negative Affectivity Schedule (Watson et al., 1988) to query participants regarding their general levels of positive and negative emotions. Although general levels of positive and negative emotion are important in their own right, our resource maintenance model would posit that more time-specific measures of emotion or mood *at work* are the drivers of productive behaviors (for a related argument, see Weiss & Cropanzano, 1996). For this reason, the meager findings in studies using

general or trait measures of PA and NA do not rule out the possibility that emotions experienced *on the job* will predict performance.

In an interesting field study, George (1991) tested this hypothesis directly. George measured PA experienced at work during the past week using a subscale from the Job Affect Scale (Burke et al., 1989). She also measured more general or trait levels of PA using the Positive Emotionality Scale from the Multidimensional Personality Questionnaire (Tellegen, 1982). Consistent with research reviewed above, George found that a trait indicator of PA was unrelated to worker altruism and customer service ratings. However, the PA one experienced at work over the past week was correlated with both altruism and customer service. Wright and Cropanzano (1999) found similar results in a later study. That is, although neither positive trait, negative trait, or positive mood today were related to performance, negative mood today was so related. These findings suggest that time- and work-specific measures of NA and PA might prove better predictors of job performance than more general or trait measures. For this reason, it is premature to close the door to future inquiry, as there are only a few studies that have examined the relation between NA-PA and job performance. We would call for future research on this topic. In addition, there is another reason why inquiry into NA and PA should not be abandoned. There is indirect evidence that NA and PA might be related to performance if we conceptualize the predictors more broadly as either extraversion or emotional stability. It is to this evidence that we now turn.

Extraversion and emotional stability. As discussed by George (1996), NA is closely related to the broad personality dimension of emotional stability (also called neuroticism), whereas PA is related to the personality dimension of extraversion (Watson, Clark, McIntyre, & Hamaker, 1992). Given this fact,

Costa and McCrae (1980) maintained that a "happy" person can be viewed as one who is high on extraversion and low on neuroticism. In keeping with the happy-productive worker thesis, it therefore follows that extraversion and emotional stability (or neuroticism) could be related to job performance. In fact, meta-analytic evidence suggests that this is probably the case (Hough, Eaton, Dunnette, Kamp, & McCloy, 1990).

Hough et al. (1990) reviewed validity coefficients among several military samples and found that emotional stability predicted three dimensions of performance (effort and leadership, personal discipline, and physical fitness and military bearing) but not two others (technical proficiency and general soldiering). Unfortunately for our purposes here, Hough and her colleagues did not examine a pure extraversion measure that we can be certain included PA. In another meta-analysis, Barrick and Mount (1991) found that emotional stability was not consistently related to performance. However, Barrick and Mount cautioned that these findings might be due to range restriction—those who are very unstable are likely to have opted out of the workforce or, at least, out of particular jobs. Extraversion was related to performance, though only for certain jobs with a strong component involving social interaction. Salgado (1997) conducted a meta-analysis of personality and performance among studies involving the European community. He found that emotional stability was a valid predictor of performance, whereas extraversion was valid among occupations with a strong social component. These findings lend limited support to the happy-productive worker thesis.

A more controversial meta-analysis was conducted by Tett, Jackson, and

Rothstein (1991), who found that extraversion was not an especially useful predictor of job performance (mean corrected validity = .16), though emotional stability was. In a critique of this study, Ones, Mount, Barrick, and Hunter (1994) raised concerns about the data analytic techniques used by Tett and his colleagues. Fortunately, a more thorough re-analysis reported by Tett, Jackson, Rothstein, and Reddon (1994) showed only small changes in the results and no major changes in the overall conclusions.

Although the meta-analytic literature is not perfectly consistent, it strongly suggests that NA (via emotional stability) and PA (via extraversion) could be related to job performance. However, this evidence is at best indirect. Although PA is correlated with extraversion and NA with emotional stability (or neuroticism), extraversion and emotional stability are substantially broader constructs than PA and NA. Whereas emotional stability or neuroticism is conceptually close to the concept of NA (for evidence, see Watson & Clark, 1984), differences remain (Costa & McCrae, 1988; McCrae & Costa, 1987). Thus, we cannot be certain whether it is the emotional aspects of extraversion and emotional stability that cause the observed differences in job performance. This meta-analytic research is promising, and we strongly recommend continued inquiry.

The Unrotated Solution: Happiness as (the Lack of) Emotional Exhaustion and Psychological Well-Being

Happiness as emotional activation-intensity: Emotional exhaustion. Somewhat more supportive findings come when researchers try to predict job performance from the other two dimensions: arousal and well-being. We shall begin with the arousal

dimension that is represented by the horizontal axis in Figure 1. As one can see, this dimension designates the intensity or vigor with which an emotion is felt. This is, of course, without regard to the hedonic tone of the emotion—one can feel fervently good or fervently bad. Although activation–intensity is an important topic in the personality literature (see Larsen & Diener, 1992, for a review), it has received minimal attention in the organizational sciences.

One exception to this generalization can be found in research on emotional exhaustion. Emotional exhaustion is widely considered to be a key component of burnout (Cordes & Dougherty, 1993; Shirom, 1989; Wright & Cropanzano, 1998). When individuals are placed under intensely stressful conditions, they experience an emotional numbness (Maslach, 1982; Maslach & Leiter, 1997). As Wright and Bonett (1997b) have pointed out, high emotional exhaustion would seem to be a manifestation of (very) low emotional arousal. Emotional exhaustion has also proven to be a useful predictor of various stress outcomes and work attitudes (Lee & Ashforth, 1996). However, until recently, the relation between emotional exhaustion and job performance has seldom been investigated.

In one study, Quattrochi-Tubin, Jones, and Breedlove (1982) found that self-reported emotional exhaustion was related to self-reported performance among social service workers. However, this work is limited in that it used self-report measures of both independent and dependent variables: exhaustion and performance. Research by Garden (1991) suggested that burnout shows a higher relation to self-reported performance than to other performance measures. In an extension of previous work, Wright and Bonett (1997a) found that emotional exhaustion was negatively correlated with supervisory ratings of job performance. However, a second study by these authors failed to

replicate this finding (Wright & Bonett, 1997b). Finally, additional supportive results were obtained by both Jones and Best (1995) and Wright and Cropanzano (1998), who found that emotional exhaustion predicted supervisory ratings of job performance. Wright and Cropanzano further observed that exhaustion scores were associated with employee voluntary turnover as much as one year later. The net result is that emotional exhaustion holds considerable promise as a predictor of job performance. Despite this promise, there are two obvious problems with using emotional exhaustion as an indicator of happiness. One is specific to emotional exhaustion, and the other is a general problem with activation–intensity measures.

Let us begin with the problem specific to emotional exhaustion. By definition, emotional exhaustion refers to the emotional numbing that accompanies burnout. Although this would seem to be indicative of low emotional intensity, an examination of the survey items suggests that this might not be the case. The most widely used measure of emotional exhaustion is a subscale from the Maslach Burnout Inventory (Maslach & Jackson, 1986). In this scale, emotional exhaustion is indicated by such items as “Working with people all day is really a strain for me,” “I feel emotionally drained from my work,” “I feel used up at the end of the workday,” “I feel frustrated with my job,” and “I feel like I’m at the end of my rope.” These items could be more indicative of the absence of positive emotion or the presence of negative emotion (or both), rather than the absence of emotion in general (cf. Meyer & Shack, 1989; Watson & Clark, 1984). If so, then this operationalization of emotional exhaustion might suggest low PA, high NA, or low well-being, rather than low activation–intensity. Further validation of the emotional exhaustion scale, particularly noting its discriminate validity to other measures of affective dispositions, is necessary to rule out this possibility. Of course, if emotional ex-

haustion proved to be an indicator of NA, PA, or well-being, then the findings we have reviewed would still support the happy-productive worker thesis, but they would not be supportive of the notion that activation-intensity is related to job performance.

Regardless of the outcomes of future research, there is a more fundamental problem with using any measure of activation-intensity as an indicator of happiness. With respect to emotion, by definition, an intensity measure does not imply either positive or negative feeling states (Larsen & Diener, 1992). It merely implies that one is experiencing some emotion with vigor; it does not denote whether that emotion is positive or negative. By all accounts, a happy person is relatively free of negative emotion, while prone to experience positive emotion (Larsen & Diener, 1992). As a consequence, activation-intensity can be ruled out as a type of happiness, though it will no doubt remain important for other purposes. We now examine happiness considered as psychological well-being.

Happiness as psychological well-being. The solid vertical line in Figure 1 captures the construct of psychological or subjective well-being. Well-being captures both positive and negative emotional states on a single axis. The high pole is anchored by such adjectives as "joyous." The low pole is anchored by "sad" and "annoyed." Thus, to be high on well-being is to be simultaneously low on negative emotion and high on positive emotion. For these reasons, various scholars have more or less equated well-being with happiness (e.g., Diener, 1984; Myers, 1993).

In support of the happy-productive worker thesis, a growing body of empirical research has found significant associations between various measures of well-being and measures of job-related performance. For example, in an experimental study, Staw and Barsade (1993) found that students who were high on well-being were superior deci-

sion makers, showed better interpersonal behaviors, and received higher overall performance ratings. Staw and Barsade's study is important for two reasons. First, it used objective, quantifiable indices of performance (e.g., an "in-basket" measure). This argues against the possibility that correlations between well-being and job performance are simply misperceptions (Robbins & DeNisi, 1994). Second, Staw and Barsade's experimental data suggest a causal relation, such that performance increases when well-being is high. This is consistent with our resource maintenance model, though it is not clear from this evidence alone whether these findings will generalize outside a laboratory setting. These generalizability concerns were addressed in a later field study by Staw et al. (1994). In their longitudinal study, Staw and his colleagues first assessed well-being. To do so, the researchers constructed a scale ad hoc out of items indicating the presence of negative emotion (e.g., depression) and the absence of positive emotion. Eighteen months later they measured various criterion variables. To be more specific, when compared with those who were low in well-being, workers high in well-being had superior performance evaluations and higher pay. Thus, well-being predicted both a subjective and an objective indicator of performance.

The work of Staw and his colleagues suggests that well-being causes increases in job performance. Also supporting this possibility is a 2-year longitudinal study by Wright, Bonett, and Sweeney (1993). Using the 8-item Index of Psychological Well-Being developed by Berkman (1971a, 1971b), Wright and his colleagues found that average well-being (Time 1 and Time 2 together) was positively related to supervisory performance ratings at Time 3. Similar results were found by Cropanzano and Wright (1999). Wright and Staw (1999) conducted

additional research to address the possibility of a causal path from well-being to performance. To that end, Wright and Staw's two longitudinal studies—incorporating multiple measures of both performance and well-being—demonstrated that psychological well-being significantly predicted supervisory performance ratings beyond the variance accounted for by prior supervisory performance ratings. These findings support the possibility that well-being is a cause of job performance.

It is also useful to ascertain the effects of other possible confounding variables. This was done in a longitudinal study by Wright and Bonett (1997b). In this study, the authors found that well-being was a positive predictor of job performance. Moreover, this relation remained significant even after controlling for the simultaneous effect of emotional exhaustion. A potentially pernicious third variable is job satisfaction. Given that job satisfaction and well-being are correlated (e.g., Diener et al., 1999; Judge & Locke, 1993), it could be that psychological well-being and job performance are only associated due to a spurious relation with job satisfaction. On the basis of the limited evidence to date, this seems not to be the case. In a cross-sectional study, Wright and Cropanzano (1997; Study 1) found that well-being was positively related to job performance beyond job satisfaction. These findings were replicated and extended in a longitudinal follow-up. In their Study 2, Wright and Cropanzano found that well-being and job performance were positively associated even when the effects of job satisfaction, NA, and PA were all taken into account. Finally, well-being remained significantly related to performance even when controlling for employee age, gender, job tenure, and educational level (Wright & Staw, 1999, Study 2).

The available data point to a common conclusion: When happiness is operationalized as well-being, it is posi-

tively related to various measures of job performance. This seems to be the case regardless of whether the criterion variables are objective indices or subjective ratings (Staw & Barsade, 1993; Staw et al., 1994). The effect also holds in experimental, cross-sectional, and longitudinal studies, even after controlling for the effects of possible confounding variables (e.g., Wright & Cropanzano, 1997; Wright & Staw, 1999). Taken together, these results are indicative of a plausible causal relation.

Directions for Future Research: Management Versus Selection Approaches to Increasing Happiness

Through this review, we have alluded to various possibilities for future research inquiry. In this final section, however, we would like to raise what we see as a pivotal research question: What are organizations to do with these findings? Assuming the findings reported in this review are replicated and supported by future inquiry, they suggest two practical possibilities. Organizations could undertake various strategies to better manage their employees to produce greater happiness, or they could select employees who are happier to begin with (Cropanzano & Wright, 1999). Little research exists comparing the efficacy of these two strategies. However, on the basis of the available literature as well as ethical considerations, we suggest that the management approach is likely to be the most beneficial.

The Management Approach to Promoting Happiness

There is some measure of stability to self-reported happiness; people who are happy (or unhappy) at one time are likely to be happy (or unhappy) at another (Diener et al., 1999; Cropanzano

& Wright, 1999). Nevertheless, this does not necessarily imply that the stability is entirely due to personal characteristics. Happiness may be stable because one's life circumstances are stable as well. For example, a person may remain at the same job or at a very similar job. For this reason, the fact of stability cannot be used to argue against the possibility of meaningful interventions designed to promote worker happiness. As of yet, the evidence for this possibility is tentative, but it is highly suggestive. In two separate longitudinal studies, Kohn and Schooler (1982) and Wright and Bonett (1992) found that employees' mental health changed over time, depending on their working conditions. Those who were employed in favorable environments reported more positive emotions than those who were in less favorable environments.

A somewhat different perspective on this problem has been taken by Judge and Locke (1993) and Judge, Locke, and Durham (1997). These researchers have asked why a person might experience feelings of high or low well-being. Judge and his colleagues argued that people make different core evaluations regarding their lives and their jobs. On the basis of these core evaluations, individuals experience certain levels of job satisfaction and well-being. For example, some workers overgeneralize failure or hold themselves to impossibly high standards. These individuals are apt to experience low job satisfaction and poor well-being. This approach offers useful guidelines for implementing changes in core evaluations to promote enhanced well-being and subsequent performance.

The Selection Approach to Promoting Happiness

Another way to raise the mean level of workforce happiness is to use some psy-

chological measure as a screening tool (Cropanzano & Wright, 1999). This process raises the mean level of workplace happiness by denying unhappy people jobs. To our thinking, this approach raises serious ethical concerns. For example, the failure to select prospective employees on the basis of their psychological well-being could depress these individuals further, which in turn could make these job candidates even more unemployable in the future. As a consequence, careful consideration of these and other related issues is of paramount importance for management personnel, their employing organizations, and practicing consultants, using various measures of psychological well-being to select happy workers.

Even if one does not share our ethical reservations, there remain practical problems in need of future research. If well-being measures were used for employee selection, then it is likely that job applicants would be motivated to fake their responses (Cropanzano & Wright, 1999). Some research suggests that those who dissimulate have a better chance of receiving offers than those who respond honestly (Douglas, McDaniel, & Snell, 1996; Rosse, Stecher, Miller, & Levin, 1998). Obviously, such faking is unlikely to maximize the utility of a staffing system.

Concluding Thoughts

In this review, we proposed that the ambiguous findings to date regarding the happy-productive worker thesis can be considered a result of the various ways in which happiness has been operationalized. Although various mechanisms exist, we offered the circumplex framework as a potentially useful taxonomy for helping to better categorize and measure happiness. Research is now needed to use systematically this taxonomy to investigate further the role of happiness-unhappiness in the prediction of performance. For instance, research has

demonstrated that performance is related to various measures of depression (Staw et al., 1994). However, although various forms of depression (e.g., fatigued vs. agitated) are similar constructs because they possess large measures of unpleasantness, they also differ because they both possess differing measures of activation (Larsen & Diener, 1992; Wright & Doherty, 1998). Awareness of this distinction could be very relevant in better predicting job performance across different job situations. As an example, although experienced levels of enthusiasm and excitement might be more influential in predicting performance or success in sales positions, experienced levels of fatigue and exhaustion might be more predictive in various client contact intensive positions, such as human services and social welfare.

We close by emphasizing an important issue. That is, this need to consider more carefully such taxonomies as the circumplex model has both theoretical and applied relevance in today's society. To be more specific, this review was primarily concerned with the relations among job satisfaction, dispositional affect, emotional exhaustion, psychological well-being, and performance. However, it should in no way be construed to suggest that job-related performance is the only, or even the most important, reason that happiness is important. In fact, too often in applied research, the emphasis appears to have been on what one might call the disease model (Wright & Cropanzano, 2000)—that is, a focus primarily concerned with fixing what's wrong with someone as opposed to developing what's right.

This extreme emphasis on the negative, to the relative neglect of the positive, has apparently had a significant and detrimental effect on how many people have come to perceive their world. For example, Myers and Diener (1995) noted that psychological publications focusing on the negative aspects of well-being outnumber their posi-

tive counterparts by a ratio of 17 to 1! Furthermore, a number of prominent researchers in the occupational health field propose that only a minority of Americans can be considered truly happy (cf. Myers & Diener, 1997). Thus, it seems evident to us that promoting happiness-related issues is an intrinsic good for which all should work (Ruack, 1999). The health model approach emphasizes the development of what's right with people (Wright & Cropanzano, 2000). If this health approach promotes better performance, and our review suggests that this is so, then so much the better. Regardless, the pursuit of employee happiness remains valuable for its own sake.

References

- Argyle, M. (1987). *The experience of happiness*. London: Methuen.
- Barrick, M. R., & Mount, M. K. (1991). The Big Five personality dimensions and job performance: A meta-analysis. *Personnel Psychology, 44*, 1-26.
- Berkman, P. L. (1971a). Measurement of mental health in a general population survey. *American Journal of Epidemiology, 94*, 105-111.
- Berkman, P. L. (1971b). Life stress and psychological well-being: A replication of Langer's analysis in the midtown Manhattan study. *Journal of Health and Social Behavior, 12*, 35-45.
- Bolger, N., & Schilling, E. A. (1991). Personality and the problems of everyday life: The role of neuroticism in exposure and reactivity to daily stressors. *Journal of Personality, 59*, 355-386.
- Brayfield, A. H., & Crockett, W. H. (1955). Employee attitudes and employee performance. *Psychological Bulletin, 52*, 396-424.
- Brief, A. P. (1998). *Attitudes in and around organizations*. Thousand Oaks, CA: Sage.
- Brief, A. P., Burke, M. J., George, J. M., Robinson, B., & Webster, J. (1988). Should negative affectivity remain an unmeasured variable in the study of job stress? *Journal of Applied Psychology, 73*, 193-198.
- Brief, A. P., Butcher, A. H., & Roberson, L. (1995). Cookies, disposition, and job attitudes: The effects of positive mood-inducing events and negative affectivity on job satisfaction in a field experiment. *Organizational Behavior and Human Decision Processes, 62*, 55-62.
- Brief, A. P., & Roberson, L. (1989). Job attitude organization: An exploratory study. *Journal of*

- Applied Social Psychology*, 19, 717–727.
- Burke, M. J., Brief, A. P., & George, J. M. (1993). The role of negative affectivity in understanding relations between self-reports of stressors and strains: A comment on the applied psychology literature. *Journal of Applied Psychology*, 78, 402–412.
- Burke, M. J., Brief, A. P., George, J. M., Roberson, L., & Webster, J. (1989). Measuring affect at work: Confirmatory analyses of competing mood structures with conceptual linkages to cortical regulatory systems. *Journal of Personality and Social Psychology*, 57, 1091–1102.
- Cascio, W. F. (1991). *Costing human resources: The financial impact of behavior in organizations* (3rd ed.). Boston: PWS-Kent.
- Cherrington, D. J., Reitz, H. J., & Scott, W. E. (1971). Effects of contingent and non-contingent reward on the relationship between satisfaction and task performance. *Journal of Applied Psychology*, 55, 531–536.
- Cordes, C. L., & Dougherty, T. W. (1993). A review and integration of research on job burnout. *Academy of Management Review*, 18, 621–656.
- Costa, P. T., & McCrae, R. R. (1980). Influence of extroversion and neuroticism on subjective well-being: Happy and unhappy people. *Journal of Personality and Social Psychology*, 33, 668–678.
- Costa, P. T., & McCrae, R. R. (1988). From catalog to classification: Murray's needs and the Five-Factor model. *Journal of Personality and Social Psychology*, 55, 258–265.
- Cropanzano, R., James, K., & Konovsky, M. A. (1993). Dispositional affectivity as a predictor of work attitudes and job performance. *Journal of Organizational Behavior*, 14, 595–606.
- Cropanzano, R., & Wright, T. A. (1999). A five-year study of change in the relationship between well-being and job performance. *Consulting Psychology Journal: Practice and Research*, 51, 252–265.
- Dember, W. N., & Brooks, J. (1989). A new instrument for measuring optimism and pessimism: Test-retest reliability and relations with happiness and religious commitment. *Bulletin of the Psychonomic Society*, 27, 365–366.
- Derryberry, D., & Reed, M. A. (1994). Temperament and attention: Orienting toward and away from positive and negative signals. *Journal of Personality and Social Psychology*, 68, 1128–1139.
- Diener, E. (1984). Subjective well-being. *Psychological Bulletin*, 95, 542–575.
- Diener, E. (1994). Assessing subjective well-being: Progress and opportunities. *Social Indicators Research*, 31, 103–157.
- Diener, E., & Larsen, R. J. (1993). The experience of emotional well-being. In M. Lewis & J. M. Haviland (Eds.), *Handbook of emotions* (pp. 404–415). New York: Guilford Press.
- Diener, E., Sandvik, E., Pavot, W., & Fujita, F. (1992). Extraversion and subjective well-being in a U.S. national probability sample. *Journal of Research in Personality*, 26, 205–215.
- Diener, E., Sandvik, E., Seidlitz, L., & Diener, M. (1993). The relationship between income and subjective well-being: Relative or absolute? *Social Indicators Research*, 28, 195–223.
- Diener, E., Suh, E. M., Lucas, R. E., & Smith, H. L. (1999). Subjective well-being: Three decades of progress. *Psychological Bulletin*, 125, 276–302.
- Douglas, E. F., McDaniel, M. A., & Snell, A. F. (1996, August). *The validity of non-cognitive measures decays when applicants fake*. Paper presented at the 1996 meeting of the Academy of Management, Cincinnati, OH.
- Fisher, C. D. (1980). On the dubious wisdom of expecting job satisfaction to correlate with performance. *Academy of Management Review*, 5, 607–612.
- Garden, A.-H. (1991). Relationship between burnout and performance. *Psychological Reports*, 68, 963–977.
- George, J. M. (1991). State or trait: Effects of positive mood on prosocial behaviors at work. *Journal of Applied Psychology*, 76, 299–307.
- George, J. M. (1992). The role of personality in organizational life: Issues and evidence. *Journal of Management*, 18, 185–210.
- George, J. M. (1996). Trait and state affect. In K. M. Murphy (Ed.), *Individual differences in behavior in organizations* (pp. 145–171). San Francisco: Jossey-Bass.
- Headey, B., & Wearing, A. (1988). The sense of relative superiority—Central to well-being. *Social Indicators Research*, 20, 497–516.
- Herzberg, F., Mausner, B., & Snyderman, B. (1959). *The motivation to work* (2nd ed.). New York: Wiley.
- Hobfoll, S. E. (1988). *The ecology of stress*. New York: Hemisphere.
- Hobfoll, S. E. (1989). Conservation of resources: A new attempt at conceptualizing stress. *American Psychologist*, 44, 513–525.
- Hobfoll, S. E. (1998). *Stress, culture, and community: The psychology and philosophy of stress*. New York: Plenum.
- Hough, L. M., Eaton, N. K., Dunnette, M. D., Kamp, J. D., & McCloy, R. A. (1990). Crite-

- tion-related validities of personality constructs and the effect of response distortion on those validities. *Journal of Applied Psychology*, 75, 581–595.
- Houser, J. D. (1927). *What the employee thinks*. Cambridge, MA: Harvard University Press.
- Iaffaldano, M. T., & Muchinsky, P. M. (1985). Job satisfaction and job performance: A meta analysis. *Psychological Bulletin*, 97, 366–373.
- Jones, R. G., & Best, R. G. (1995, August). *Further examination of the nature and impact of emotional work requirements*. Paper presented at the annual meeting of the Academy of Management, Vancouver, British Columbia, Canada.
- Judge, T. A. (1992). The dispositional perspective in human resources management. In G. R. Ferris & K. M. Rowland (Eds.), *Research in personnel and human resources management* (Vol. 10, pp. 31–72). Greenwich, CT: JAI Press.
- Judge, T. A., Hanisch, K. A., & Drankoski, R. D. (1995). Human resources management and employee attitudes. In G. R. Ferris, S. D. Rosen, & D. T. Barnum (Eds.), *Handbook of human resources management* (pp. 574–596). Oxford, England: Blackwell Publishers.
- Judge, T. A., & Locke, E. A. (1993). Effects of dysfunctional thought processes on subjective well-being and job satisfaction. *Journal of Applied Psychology*, 78, 475–490.
- Judge, T. A., Locke, E. A., & Durham, C. C. (1997). The dispositional causes of job satisfaction: A core evaluations approach. In B. M. Staw & L. L. Cummings (Eds.), *Research in Organizational Behavior* (Vol. 17, pp. 151–188). Greenwich, CT: JAI Press.
- Judge, T. A., Thoresen, C. J., Bono, J. E., & Patton, G. K. (1998, August). *The job satisfaction–job performance relationship: 1939–1998*. Paper presented at the annual meeting of the Academy of Management, San Diego, CA.
- Kohn, M. L., & Schooler, C. (1982). Job conditions and personality: A longitudinal assessment. *American Journal of Sociology*, 87, 1257–1286.
- Landy, F. W. (1985). *The psychology of work behavior* (3rd ed.). Homewood, IL: Dorsey Press.
- Larsen, R. J., & Diener, E. (1992). Promises and problems with the circumplex model of emotion. *Review of Personality and Social Psychology*, 13, 25–59.
- Larsen, R. J., & Ketelaar, T. (1989). Extraversion, neuroticism, and susceptibility to positive and negative mood induction procedures. *Personality and Individual Differences*, 10, 1221–1228.
- Larsen, R. J., & Ketelaar, T. (1991). Personality and susceptibility to positive and negative emotional states. *Journal of Personality and Social Psychology*, 61, 132–140.
- Lawler, E. E., & Porter, L. W. (1967). The effects of performance on job satisfaction. *Industrial Relations*, 7, 20–28.
- Lee, R. T., & Ashforth, B. E. (1996). A meta-analytic examination of the correlates of the three dimensions of job burnout. *Journal of Applied Psychology*, 81, 123–133.
- Levin, I., & Stokes, J. P. (1989). Dispositional approach to job satisfaction: Role of negative affectivity. *Journal of Applied Psychology*, 74, 752–758.
- Locke, L. A. (1976). The nature and causes of job satisfaction. In M. D. Dunnette (Ed.), *Handbook of industrial and organizational psychology* (1st ed., pp. 1297–1349). Chicago, IL: Rand McNally.
- Maslach, C. (1982). *Burnout: The cost of caring*. Englewood Cliffs, NJ: Prentice Hall.
- Maslach, C., & Jackson, S. E. (1986). *Maslach Burnout Inventory* (2nd ed.). Palo Alto, CA: Consulting Psychologists Press.
- Maslach, C., & Leiter, M. P. (1997). *The truth about burnout: How organizations cause personnel stress and what to do about it*. San Francisco: Jossey-Bass.
- Mattox, W. R., Jr. (1999, April 8). The great (clinical) depression: Dow 10K shows money can't buy happiness. *San Francisco Chronicle*, p. A23.
- Mayer, J. D., & Gaschke, Y. N. (1988). The experience and meta-experience of mood. *Journal of Personality and Social Psychology*, 55, 102–111.
- McCrae, R. R., & Costa, P. T., Jr. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology*, 52, 81–90.
- Michalos, A. C. (1985). Multiple discrepancies theory (MDT). *Social Indicators Research*, 16, 347–413.
- Meyer, G. J., & Shack, J. R. (1989). Structural convergence of mood and personality: Evidence for old and new directions. *The Journal of Personality and Social Psychology*, 57, 691–706.
- Myers, D. G. (1993). *The pursuit of happiness*. New York: Avon Books.
- Myers, D. G., & Diener, E. (1995). Who is happy? *Psychological Science*, 6, 10–19.
- Myers, D. G., & Diener, E. (1997). The new pursuit of happiness. *The Harvard Medical Health Letter*, 14, 4–7.

- Nord, W. R. (1976). Attitudes and performance. In W. R. Nord (Ed.), *Concepts and controversies in organizational behavior* (pp. 518–525). Glenview, IL: Scott, Foresman.
- Ones, D. S., Mount, M. K., Barrick, M. R., & Hunter, J. E. (1994). Personality and job performance: A critique of Tett, Jackson, & Rothstein (1994) meta-analysis. *Personnel Psychology*, 47, 147–171.
- Parducci, A. (1995). *Happiness, pleasure, and judgment: The contextual theory and its applications*. Hillsdale, NJ: Erlbaum.
- Pavot, W., Diener, E., Colvin, C. R., & Sandvik, E. (1991). Further validation of the Satisfaction With Life Scale: Evidence for cross-method convergence of well-being measures. *Journal of Personality Assessment*, 57, 149–161.
- Petty, M. M., McGee, G. W., & Cavender, J. W. (1984). A meta-analysis of the relationship between individual job satisfaction and individual performance. *Academy of Management Review*, 9, 712–721.
- Podsakoff, P. M., & Williams, L. J. (1986). The relationship between job performance and job satisfaction. In E. A. Locke (Ed.), *Generalizing from laboratory to field studies* (pp. 207–245). Lexington, MA: Lexington Books.
- Quattrochi-Tubin, S., Jones, J. W., & Breedlove, V. (1982). The burnout syndrome in geriatric counselors and service workers. *Activities, Adaptation, and Aging*, 3, 65–76.
- Robbins, T. L., & DeNisi, A. S. (1994). A closer look at interpersonal affect as a distinct influence on cognitive processing in performance evaluations. *Journal of Applied Psychology*, 79, 341–453.
- Roethlisberger, F. J., & Dickson, W. J. (1939). *Management and the worker*. Cambridge, MA: Harvard University Press.
- Rosse, J. G., Stecher, M. D., Miller, J. L., & Levin, R. A. (1998). The impact of response distortion on preemployment personality testing and hiring decisions. *Journal of Applied Psychology*, 83, 634–644.
- Ruack, J. K. (1999, February 12). Redefining the good life: A new focus in the social sciences. *The Chronicle of Higher Education*, pp. A13–A15.
- Russell, J. A. (1978). Evidence of convergent validity on the dimensions of affect. *Journal of Personality and Social Psychology*, 36, 1152–1168.
- Russell, J. A. (1979). Affective space is bipolar. *Journal of Personality and Social Psychology*, 37, 345–356.
- Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39, 1161–1178.
- Russell, J. A., Lewicka, M., & Nitt, T. (1989). A cross-cultural study of a circumplex model of affect. *Journal of Personality and Social Psychology*, 57, 848–856.
- Russell, J. A., & Ridgeway, D. (1983). Dimensions underlying children's emotion concepts. *Developmental Psychology*, 19, 795–804.
- Russell, J. A., Weiss, A., & Mendelsohn, G. A. (1989). Affect Grid: A single-item scale of pleasure and arousal. *Journal of Personality and Social Psychology*, 57, 493–502.
- Rusting, C. G., & Larsen, R. J. (1997). Extraversion, neuroticism, and susceptibility to positive and negative affect: A test of two theoretical models. *Personality and Individual Differences*, 22, 607–612.
- Salgado, J. F. (1997). The five factor model of personality and job performance in the European Community. *Journal of Applied Psychology*, 82, 30–43.
- Sandvik, E., Diener, E., & Seidlitz, L. (1993). Subjective well-being: The convergence and stability of self-report measures. *Journal of Personality*, 61, 317–342.
- Schwab, D. P., & Cummings, L. L. (1970). Theories of performance and satisfaction: A review. *Industrial Relations*, 9, 408–430.
- Seidlitz, L., & Diener, E. (1993). Memory for positive versus negative events: Theories for the differences between happy and unhappy persons. *Journal of Personality and Social Psychology*, 64, 654–664.
- Seidlitz, L., Wyer, R. S., & Diener, E. (1997). Cognitive correlates of subjective well-being: The processing of valenced life events by happy and unhappy persons. *Journal of Research in Personality*, 31, 240–256.
- Seligman, M. E. P. (1991). *Learned optimism*. New York: Random House.
- Seligman, M. E. P. (1994). *What you can change and what you can't*. New York: Knopf.
- Seligman, M. E. P. (1995). The effectiveness of psychotherapy: The *Consumer's Report* study. *American Psychologist*, 50, 965–974.
- Shirom, A. (1989). Burnout in work organizations. In C. L. Cooper & I. Robertson (Eds.), *International review of industrial and organizational psychology* (pp. 25–48). New York: Wiley.
- Staw, B. M. (1986). Organizational psychology and the pursuit of the happy/productive worker. *California Management Review*, XXVIII(4), 40–53.
- Staw, B. M., & Barsade, S. G. (1993). Affect and

- managerial performance: A test of the sadder-but-wiser vs. happier-and-smarter hypotheses. *Administrative Science Quarterly*, 38, 304–331.
- Staw, B. M., Sutton, R. I., & Pelled, L. H. (1994). Employee positive emotion and favorable outcomes at the workplace. *Organization Science*, 5, 51–71.
- Sullivan, J. J. (1989). Self theories and employee motivation. *Journal of Management*, 15, 345–363.
- Tellegen, A. (1982). *Brief manual for the Differential Personality Questionnaire*. Unpublished manuscript, University of Minnesota.
- Tett, R. P., Jackson, D. N., & Rothstein, M. (1991). Personality measures as predictors of job performance: A meta-analytic review. *Personnel Psychology*, 44, 703–742.
- Tett, R. P., Jackson, D. N., Rothstein, M., & Reddon, J. R. (1994). Meta-analysis of personality–job performance relations: A reply to Ones, Mount, Barrick, and Hunter (1994). *Personnel Psychology*, 47, 157–172.
- Veenhoven, R. (1988). The utility of happiness. *Social Indicators Research*, 20, 333–354.
- Vroom, V. H. (1964). *Work and motivation*. New York: Wiley.
- Warr, P. (1987). *Work, employment, and mental health*. New York: Oxford University Press.
- Warr, P. (1990). The measurement of well-being and other aspects of mental health. *Journal of Occupational Psychology*, 63, 193–210.
- Watson, D. (1988). Intraindividual and interindividual analysis of positive and negative affect: Their relation to health complaints, perceived stress, and daily activities. *Journal of Personality and Social Psychology*, 54, 1020–1030.
- Watson, D., & Clark, L. A. (1984). Negative affectivity: The disposition to experience negative emotional states. *Psychological Bulletin*, 96, 465–490.
- Watson, D., Clark, L. A., & McIntyre, C. W., & Hamaker, S. (1992). Affect, personality, and social activity. *Journal of Personality and Social Psychology*, 63, 1011–1025.
- Watson, D., Clark, L. A., & Tellegen, A. (1984). Cross-cultural convergence in the structure of mood: A Japanese replication and a comparison with U.S. findings. *Journal of Personality and Social Psychology*, 47, 127–144.
- Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS Scales. *Journal of Personality and Social Psychology*, 54, 1063–1070.
- Watson, D., & Pennebaker, J. W. (1989). Health complaints, stress, and distress: Exploring the central role of negative affectivity. *Psychological Review*, 96, 234–254.
- Watson, D., & Slack, A. K. (1993). General factors of affective temperament and their relation to job satisfaction over time. *Organizational Behavior and Human Decision Processes*, 54, 181–202.
- Watson, D., & Tellegen, A. (1985). Toward the structure of affect. *Psychological Bulletin*, 98, 219–235.
- Weiss, H. M., & Cropanzano, R. (1996). An affective events approach to job satisfaction. In B. M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (Vol. 18, pp. 1–74). Greenwich, CT: JAI Press.
- Wright, T. A. (1999). An eight-step approach to enhance the application of organizational research. In L. Larwood & U. E. Gattiker (Eds.), *Impact analysis—How research enters application and makes a difference* (pp. 137–143). Hillsdale, NJ: Erlbaum.
- Wright, T. A., & Bonett, D. G. (1992). The effect of turnover on work satisfaction and mental health: Support for a situational perspective. *Journal of Organizational Behavior*, 13, 603–615.
- Wright, T. A., & Bonett, D. G. (1997a). The contribution of burnout to work performance. *Journal of Organizational Behavior*, 18, 491–499.
- Wright, T. A., & Bonett, D. G. (1997b). The role of pleasantness and activation-based well-being in performance prediction. *Journal of Occupational Health Psychology*, 2, 212–219.
- Wright, T. A., Bonett, D. G., & Sweeney, D. A. (1993). Mental health and work performance: Results of a longitudinal field study. *Journal of Occupational and Organizational Psychology*, 66, 277–284.
- Wright, T. A., & Cropanzano, R. (1997). Well-being, satisfaction and job performance: Another look at the happy/productive worker relationship. In L. N. Dosier & J. B. Keys (Eds.), *Academy of management: Best paper proceedings* (pp. 364–368). Statesboro, GA: Office of Publications and Faculty Research Services, College of Business Administration, Georgia Southern University.
- Wright, T. A., & Cropanzano, R. (1998). Emotional exhaustion as a predictor of job performance and voluntary turnover. *Journal of Applied Psychology*, 83, 486–493.
- Wright, T. A., & Cropanzano, R. (1999). *When a happy worker is a productive worker: A preliminary examination of three models*. Manuscript submitted for publication.

- Wright, T. A., & Cropanzano, R. (2000). The role of organizational behavior in occupational health psychology: A view as we approach the millennium. *Journal of Occupational Health Psychology, 5*, 5–10.
- Wright, T. A., & Doherty, E. M. (1998). Organizational behavior 'rediscovers' the role of emotional well-being. *Journal of Organizational Behavior, 19*, 481–485.
- Wright, T. A., & Staw, B. M. (1999). Affect and favorable work outcomes: Two longitudinal tests of the happy-productive worker thesis. *Journal of Organizational Behavior, 20*, 1–23.